THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
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ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 60 km/h	speed = distance ÷ time
2	**(d)** 60 km/h	speed = distance ÷ time
3	**(a)** 180 km	distance = speed × time
4	**(c)** 120 km	distance = speed × time
5	**(d)** 4 hours	time = distance ÷ speed
6	**(c)** 30 m/s	1 km/h = 5/18 m/s
7	**(a)** 18 km/h	multiply by 18/5
8	**(d)** length	longer pendulum takes more time
9	**(d)** clepsydra	water flow measures time
10	**(d)** uniform motion	constant speed

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Time taken for one complete oscillation (2).	2
12	Second (1); metre per second (1).	1+1
13	Distance travelled per unit time (1); speed = distance ÷ time (1).	1+1
14	Equal distances in equal times; unequal distances in equal times (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	60 km/h (1.5); 16.67 m/s (1.5).	1½+1½
16	5 hours (3).	3
17	Hang a bob and swing small angle (1); time 20 oscillations with a stopwatch (1); divide to get time period (1).	1+1+1
	OR: A gnomon casts a shadow (1.5); shadow position indicates time (1.5).	1½+1½
18	25 km (1.5); 25/3 = 8.33 km/h (1.5).	1½+1½
	OR: Equal distance in equal time (1.5); 5 m/s (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Distance/time; m/s (1). (ii) 60 km/h (2). (iii) 16.67 m/s (2).	1+2+2
	OR: (i) 1.8 s (1). (ii) 33 (1). (iii) Time period increases (1). (iv) Reduces error (2).	1+1+1+2
20	(i) Sun shadow; water flow (3). (ii) Sundial needs sun; clepsydra needs refilling (2).	3+2
	OR: (i) 180 km (1). (ii) 3 h (1). (iii) 60 km/h (1). (iv) No, unequal distances (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 50 km/h	speed = distance ÷ time
2	**(c)** 50 km/h	speed = distance ÷ time
3	**(d)** 90 km	distance = speed × time
4	**(a)** 4 hours	time = distance ÷ speed
5	**(b)** 10 m/s	1 km/h = 5/18 m/s
6	**(d)** 36 km/h	multiply by 18/5
7	**(c)** second	s
8	**(c)** regular oscillation of a pendulum	to measure time
9	**(d)** distance covered per unit time	definition
10	**(b)** non-uniform motion	changing speed

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Instrument using Sun's shadow to tell time (1); needs sunlight (1).	1+1
12	150 ÷ 3 = 50 km/h (2).	2
13	72 × 5/18 = 20 m/s (2).	2
14	An ancient water clock (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance is how far; speed is how fast (1.5); units m and m/s (1.5).	1½+1½
16	40 (3).	3
17	Needs Sun, location specific (1.5); clock works any time (1.5).	1½+1½
	OR: To give the same result everywhere (3).	3
18	25 m/s (1.5); 28.8 km/h (1.5).	1½+1½
	OR: 8 m/s (1.5); 28.8 km/h (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Definitions and examples (2). (ii) Straight line; curve (2). (iii) For varying speeds (1).	2+2+1
	OR: (i) 30 m/s; 90 km/h (2). (ii) 108 km/h (1). (iii) 300 m (2).	2+1+2
20	(i) 8 m/s (1). (ii) 28.8 km/h (1). (iii) 50 s (1). (iv) Stopwatch (2).	1+1+1+2
	OR: (i) Pahar, ghati (1). (ii) Sundial, water clock (2). (iii) Common reference (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.